

A Proofs

A.1 Auxiliary Lemmas

We define a (mixed-strategy) strategy σ_i for agent i to be a map from priors b_i to elements of the simplex $\Delta(\{\mathcal{D}, \mathcal{I}, \mathcal{S}\})$. In others words, σ_i specifies for each ideological prior b_i of agent i the probability that she will play each of the three actions, $\mathcal{D}$, $\mathcal{I}$, and $\mathcal{S}$. We let σ_{-i} denote the (vector of) strategies of all agents other than agent i .

Lemma A.1. *Given any set of strategies σ_{-i} , agent i 's best response is a cutoff strategy with cutoffs (b_i^*, b_i^{**}) such that if $b_i < b_i^*$ agent i dislikes ($\mathcal{D}$), if $b_i^* < b_i < b_i^{**}$ agent i ignores ($\mathcal{I}$), and if $b_i > b_i^{**}$ agent i shares ($\mathcal{S}$).*

Proof of Lemma A.1. When agent i receives an article, she forms (ex-post) belief π_i about the article's veracity which depends only on the observables (r, m) . By Bayes' rule:

$$\pi_i \equiv \mathbb{P}[\nu = \mathcal{T} \mid r, m = R] = \frac{\mathbb{P}[m = R \mid r, \nu = \mathcal{T}] \mathbb{P}[\nu = \mathcal{T} \mid r]}{\mathbb{P}[m = R \mid r, \nu = \mathcal{M}] \mathbb{P}[\nu = \mathcal{M} \mid r] + \mathbb{P}[m = R \mid r, \nu = \mathcal{T}] \mathbb{P}[\nu = \mathcal{T} \mid r]}.$$

By the law of total probability, we have:

$$\begin{aligned} \mathbb{P}[m = R \mid r, \nu = \mathcal{T}] &= \mathbb{P}[m = R \mid \nu = \mathcal{T}] = \mathbb{P}[m = R \mid \theta = R, \nu = \mathcal{T}] \mathbb{P}[\theta = R] + \mathbb{P}[m = R \mid \theta = L, \nu = \mathcal{T}] \mathbb{P}[\theta = L] \\ &= pb_i + (1 - p)(1 - b_i); \\ \mathbb{P}[m = R \mid r, \nu = \mathcal{M}] &= \mathbb{P}[m = R \mid \nu = \mathcal{M}] = \mathbb{P}[m = R \mid \theta = R, \nu = \mathcal{M}] \mathbb{P}[\theta = R] + \mathbb{P}[m = R \mid \theta = L, \nu = \mathcal{M}] \mathbb{P}[\theta = L] \\ &= qb_i + (1 - q)(1 - b_i). \end{aligned}$$

Putting these together we obtain equation (2). Moreover, π_i is monotone in b_i since

$$\frac{\partial \pi_i}{\partial b_i} = \frac{(1 - \phi(r))\phi(r)(p - q)}{(1 - b_i + q(1 - \phi(r))(2b_i - 1) - p(\phi(r) - 2\phi(r)b_i))^2} > 0.$$

Note that $U_i(\mathcal{I})$ and $U_i(\mathcal{D})$ is independent of σ_{-i} , and in particular $\mathcal{D}$ is a better response to $\mathcal{I}$ if and only if $\pi_i < (\tilde{u} - \tilde{c})/\tilde{u}$. Because π_i is monotone in b_i , this implies there exists some $\tilde{b}$ where $\mathcal{D}$ is a better response to $\mathcal{I}$ if and only if $b_i < \tilde{b}$ (where $\tilde{b} = 1$ if disliking dominates ignoring and $\tilde{b} = 0$ if ignoring dominates disliking). Next, recall that the payoff to sharing is $U_i(\mathcal{S}) = U_i^{(1)} + U_i^{(2)}$, where $U_i^{(1)} = u\mathbf{1}_{\nu=\mathcal{T}} - c\mathbf{1}_{\nu=\mathcal{M}}$ and $U_i^{(2)} = \kappa S_i - dD_i$. Observe that, as before, $U_i^{(1)}$ is independent of σ_{-i} and has expected payoff $(u + c)\pi_i - c$, which is monotonically increasing in π_i . Moreover, $\mathbb{E}_{\mathbf{P}, \sigma_{-i}}[\kappa S_i - dD_i]$ does not depend on b_i . Because π_i is monotone in b_i , we see that $U_i(\mathcal{S})$ is increasing in b_i , $U_i(\mathcal{I})$ is constant in b_i (it is always zero), and $U_i(\mathcal{D})$ is decreasing in b_i (it is equal to $\tilde{u}(1 - \pi_i) - \tilde{c}$). This implies that either (i) ignoring dominates sharing, (ii) sharing dominates ignoring, or (iii) $U_i(\mathcal{S}) = 0$ for some prior b' :

- (i) If ignoring dominates sharing, we set $(b_i^*, b_i^{**}) = (\tilde{b}, 1)$.
- (ii) If sharing dominates ignoring, then either sharing dominates disliking (in which case set $(b_i^*, b_i^{**}) = (0, 0)$), disliking dominates sharing (in which case we set $(b_i^*, b_i^{**}) = (1, 1)$), or there exists some